

# AFRILANG Stdlib

## std/c/graphclique

**Français.** Bibliothèque AFRILANG 'std/c/graphclique' (niveau complexe, catégorie complexe). Elle expose 7 fonction(s) : maxCliqueSize, cliqueNumber, isClique, cliqueCover, triangleCount, clusteringCoeff, density.

```
'import "std/c/graphclique"' · 'use graphclique'
```

- 'maxCliqueSize(adj list number, n number) → number' - 'cliqueNumber(adj list number, n number) → number' - 'isClique(adj list number, n number, nodes list number) → bool' - 'cliqueCover(adj list number, n number) → number' - 'triangleCount(adj list number, n number) → number' - 'clusteringCoeff(adj list number, n number) → list number' - 'densit

**English.** AFRILANG library 'std/c/graphclique' (tier complex, category complex). Exports 7 function(s): maxCliqueSize, cliqueNumber, isClique, cliqueCover, triangleCount, clusteringCoeff, density.

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- 'maxCliqueSize(adj list number, n number) → number' - 'cliqueNumber(adj list number, n number) → number' - 'isClique(adj list number, n number, nodes list number) → bool' - 'cliqueCover(adj list number, n number) → number' - 'triangleCount(adj list number, n number) → number' - 'clusteringCoeff(adj list number, n number) → list number' - 'density(adj list nu

|                       |                       |               |
|-----------------------|-----------------------|---------------|
| Catégorie / Category  | Modules complexe (c/) |               |
| Niveau / Tier         | complexe              | June 16, 2026 |
| Fonctions / Functions | 7                     |               |

## Contents

## Français

### 1. Vue d'ensemble

Bibliothèque AFRILANG 'std/c/graphclique' (niveau complex, catégorie complex). Elle expose 7 fonction(s) : maxCliqueSize, cliqueNumber, isClique, cliqueCover, triangleCount, clusteringCoeff, density.

'import "std/c/graphclique"' · 'use graphclique'

```
- `maxCliqueSize(adj list number, n number) → number`
- `cliqueNumber(adj list number, n number) → number`
- `isClique(adj list number, n number, nodes list number) → bool`
- `cliqueCover(adj list number, n number) → number`
- `triangleCount(adj list number, n number) → number`
- `clusteringCoeff(adj list number, n number) → list number`
- `density(adj list number, n number) → number`
```

### 2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/graphclique"
use graphclique
```

Niveau : complex · Catégorie : Modules complex (c/)  
Domaines avancés – graphes, NLP, réseaux complexes.

### 3. Référence API

- maxCliqueSize(adj list number, n number) → number•
- cliqueNumber(adj list number, n number) → number•
- isClique(adj list number, n number, nodes list number) → bool•
- cliqueCover(adj list number, n number) → number•
- triangleCount(adj list number, n number) → number•
- clusteringCoeff(adj list number, n number) → list•
- density(adj list number, n number) → number

### 4. Exemples d'utilisation

```
import "std/c/graphclique"
use graphclique
```

```
create result = maxCliqueSize(/* args */)
say result
```

```
create result = cliqueNumber(/* args */)
say result
```

```
create result = isClique(/* args */)
say result
```

```
create result = cliqueCover(/* args */)
say result
```

```
create result = triangleCount(/* args */)
say result
```

```
create result = clusteringCoeff(/* args */)
say result
```

## 5. Bonnes pratiques

- Préférez ``import "std/c/graphclique"`` en tête de fichier.
- Lancez ``afriLang check`` avant de livrer.
- Combinez avec les modules `stdlib` de la même catégorie.
- Voir `/stdlib/` pour le source live et les démos playground.
- Signalez les problèmes sur GitHub si une export manque.

## 6. Annexe — source

module graphclique

```
function maxCliqueSize(adj list number, n number) returns number
end
```

```
function cliqueNumber(adj list number, n number) returns number
end
```

```
function isClique(adj list number, n number, nodes list number) returns bool
end
```

```
function cliqueCover(adj list number, n number) returns number
end
```

```
function triangleCount(adj list number, n number) returns number
end
```

```
function clusteringCoeff(adj list number, n number) returns list number
end
```

```
function density(adj list number, n number) returns number
end
end
```

## English

### 1. Overview

AFRILANG library 'std/c/graphclique' (tier complex, category complex). Exports 7 function(s): maxCliqueSize, cliqueNumber, isClique, cliqueCover, triangleCount, clusteringCoeff, density.

'import "std/c/graphclique"' · 'use graphclique'

```
- `maxCliqueSize(adj list number, n number) → number`
- `cliqueNumber(adj list number, n number) → number`
- `isClique(adj list number, n number, nodes list number) → bool`
- `cliqueCover(adj list number, n number) → number`
- `triangleCount(adj list number, n number) → number`
- `clusteringCoeff(adj list number, n number) → list number`
- `density(adj list number, n number) → number`
```

### 2. Installation & import

Add the module to your program:

```
import "std/c/graphclique"
use graphclique
```

Tier: complex · Category: Complex modules (c/)  
Advanced domains – graphs, NLP, complex networks.

### 3. API reference

- maxCliqueSize(adj list number, n number) → number•
- cliqueNumber(adj list number, n number) → number•
- isClique(adj list number, n number, nodes list number) → bool•
- cliqueCover(adj list number, n number) → number•
- triangleCount(adj list number, n number) → number•
- clusteringCoeff(adj list number, n number) → list•
- density(adj list number, n number) → number

### 4. Usage examples

```
import "std/c/graphclique"
use graphclique
```

```
create result = maxCliqueSize(/* args */)
say result
```

```
create result = cliqueNumber(/* args */)
say result
```

```
create result = isClique(/* args */)
say result
```

```
create result = cliqueCover(/* args */)
say result
```

```
create result = triangleCount(/* args */)
say result
```

```
create result = clusteringCoeff(/* args */)
say result
```

## 5. Best practices

- Prefer ``import "std/c/graphclique"`` at the top of your file.
- Run ``afriLang check`` before shipping.
- Combine with related stdlib modules from the same category.
- See `/stdlib/` for the live source and playground demos.
- Report issues on GitHub if an export is missing or undocumented.

## 6. Appendix — source

module graphclique

```
function maxCliqueSize(adj list number, n number) returns number
end
```

```
function cliqueNumber(adj list number, n number) returns number
end
```

```
function isClique(adj list number, n number, nodes list number) returns bool
end
```

```
function cliqueCover(adj list number, n number) returns number
end
```

```
function triangleCount(adj list number, n number) returns number
end
```

```
function clusteringCoeff(adj list number, n number) returns list number
end
```

```
function density(adj list number, n number) returns number
end
end
```

## Référence rapide / Quick reference

- Import : `import "std/c/graphclique"`
- Vérification : `afriLang check`
- Documentation : <https://afriLang.dev/stdlib/std/c/graphclique/>

## Source (extrait)

```
module graphclique

function maxCliqueSize(adj list number, n number) returns number
end

function cliqueNumber(adj list number, n number) returns number
end

function isClique(adj list number, n number, nodes list number) returns bool
end

function cliqueCover(adj list number, n number) returns number
end

function triangleCount(adj list number, n number) returns number
end

function clusteringCoeff(adj list number, n number) returns list number
end

function density(adj list number, n number) returns number
end
end
```